

Silverline House — 25-Storey Residential HRB

Facade Interface and Support Design

Part C — Site Preparation | BS EN 1993 / CWCT

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1 Introduction

This document describes the interface between the primary structural frame and the rainscreen cladding system, including the design of the support brackets, cleats, and fixing arrangements. The facade support system has been designed to transfer vertical dead loads and horizontal wind loads from the facade to the structural slab edge, whilst accommodating differential movement between the facade and the structure arising from temperature variation, creep, and shrinkage. The design has been carried out in accordance with BS EN 1993-1-8 (steel connections) and the recommendations of CWCT Technical Notes. All brackets are in Grade 316L stainless steel with BBA certification.

This document forms part of the Gateway 2 application for Building Control Approval submitted to the Building Safety Regulator under the Building Safety Act 2022. The information presented herein has been prepared in accordance with the CLC Guidance Suite for new Higher-Risk Buildings and the requirements set out in Guidance Note 04 (Application Information Schedule) and Guidance Note 06 (Document Management and Submission). All design work described in this document has been carried out by suitably qualified engineers and specialists with relevant experience in higher-risk residential buildings.

2 Facade System Overview

The primary rainscreen cladding system is a rear-ventilated facade with horizontal support rails at 600 mm vertical centres attached to stainless steel L-brackets fixed at 900 mm horizontal centres to the structural slab edge. The bracket design must accommodate a slab-edge to finished face dimension of 250 mm, a maximum vertical load per bracket of 3.2 kN, and a maximum horizontal (wind) load per bracket of 1.8 kN. Thermal break pads are provided between each bracket foot and the concrete slab edge to limit thermal bridging. Adjustment slots in the bracket allow for construction tolerances of ± 15 mm in all three axes. The rail-to-bracket connection is via stainless steel M12 bolts with load-spreading plates, designed to resist the combined effects of dead load, wind load, and seismic actions.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

3.1 System Self-Weight

The self-weight of the facade system has been assessed in accordance with BS EN 1991-1-1 and the manufacturer's declared product data. The assessment covers all components of the installed facade from the slab edge fixing point to the outer panel face, including all brackets, rails, fixings, insulation, air and vapour control layer, and cladding panels.

The rainscreen cladding system has a self-weight of 1.8 kN/m² including all fixings and insulation. This value is derived from the system component data confirmed by the specialist cladding subcontractor. The 1.8 kN/m² self-weight includes: fibre cement panels (0.95 kN/m²), horizontal support rails (0.18 kN/m²), stainless steel brackets and fixings (0.22 kN/m²), 120 mm mineral wool insulation (0.22 kN/m²), and air and vapour control layer (0.03 kN/m²).

This self-weight is applied to the structural slab edge as a uniformly distributed line load for structural design purposes. The line load per metre of slab edge is calculated as 1.8 kN/m² × storey height of 3.2 m = 5.76 kN/m. This value should be used in the slab edge and bracket design calculations contained in the Structural Calculation Pack (document ref 2.1.1).

3.2 Bracket Design

Each bracket is designed to carry vertical loads from the cladding panels plus self-weight of rails, insulation, and fixings above. Horizontal loads are transferred via shear at the bracket-to-slab fixing. The fixing to the slab edge uses 12 mm stainless steel anchor bolts cast into the slab edge at the time of construction; post-drill fixings are not permitted. The bracket arrangement has been checked for combined bending and shear, and for torsion arising from the offset between the centroid of the cladding panel and the face of the structural slab. All bracket calculations comply with BS EN 1993-1-8 and ETAG 001 (Anchors for use in concrete). Pull-out tests in accordance with BS 8539:2012 will be carried out on a minimum of 10% of cast-in fixings during construction.

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